

Prelim Skills Exam

Tools Needed:

1. Control Node (CN) - 1
2. Manage Node (MN) - 2 Ubuntu

Procedure:

1. Note: You are required to create a document report of the steps you will do for this exam. All screenshots should be labeled and explained properly. LABELED AND EXPLAIN EACH CODE (PLAYBOOK) No explanation = Minus Points
2. Create a repository in your GitHub account and label it as Surname_PrelimExam
3. Clone your new repository in your CN.
4. In your CN, create an inventory file and ansible.cfg files.
5. Create an Ansible playbook that does the following with an input of a config.yaml file for both Manage Nodes
 - Installs the latest python3 and pip3
 - use pip3 as default pip
 - use python3 as default python
 - Install Java open-jdk
 - Install MariaDB as well as starting the server, create a database and a table using mariaDB and input one record into a table USING ANSIBLE ONLY
 - Create Motd containing the text defined by a variable defined in config.yaml file and if there is no variable input the default motd is "Ansible Managed node by (your user name)"
 - Create a user with a variable defined in config.yaml
5. PUSH and COMMIT your PrelimExam in your GitHub repo
6. Your document report should be submitted here.
7. For your prelim exam to be counted, please paste your repository link here. (Failure to submit will result in ZERO)
8. NO USE OF EXTERNAL WEBSITES SUCH AS REDDIT, CHATGPT, GITHUB, GEMINI, AND CLAUDE. FAILURE TO COMPLY WITH RESULT IN ZERO. AI OVERVIEW SEARCH WILL ALSO RESULT IN A FAILED GRADE. Always use the "-ai" in the search bar, only use Google Search Engine.

Output:

Clone your new repository in your CN.

```
jalene@workstation: ~/Santiago_PrelimExam
jalene@workstation:~$ git clone "git@github.com:jaliebee/Santiago_PrelimExam.git"
Cloning into 'Santiago_PrelimExam'...
warning: You appear to have cloned an empty repository.
jalene@workstation:~$ cd Santiago_PrelimExam
jalene@workstation:~/Santiago_PrelimExam$
```

After I created a new repository on my github, I cloned it on my Ubuntu.

In your CN, create an inventory file and ansible.cfg files.

```
→ Santiago_PrelimExam
ansible.cfg U X inventory.ini U ! playbook.yml U
ansible.cfg
1 [defaults]
2 inventory = inventory.ini
3 private_key_file = /home/jalene/.ssh/id_ed25519
```

```
→ Santiago_PrelimExam
ansible.cfg U inventory.ini U X ! playbook.yml U
inventory.ini
1 [db_servers]
2 [web_servers]
3 server1 ansible_host=192.168.56.102 ansible_user=jalene target_user=jalene
4 server2 ansible_host=192.168.56.103 ansible_user=jalene target_user=jalene
```

The ansible.cfg and inventory.ini came from the past activities. No changes needed, IP addresses of Manage Nodes are already there.

- Installs the latest python3 and pip3

```
- name: Install Python3 and pip3
  package:
    name:
      - python3
      - python3-pip
    state: latest
```

```
TASK [Install Python3 and pip3] *****
changed: [server1]
changed: [server2]
```

This code shows how to install both Python3 and pip3 using Ansible. I installed it correctly as shown on the output.

```
jalene@server1:~$ python3 -V
Python 3.12.3
jalene@server1:~$ pip3 -V
pip 24.0 from /usr/lib/python3/dist-packages/pip (python 3.12)
```

```
jalene@server2:~$ python3 -V
Python 3.12.3
jalene@server2:~$ pip3 -V
pip 24.0 from /usr/lib/python3/dist-packages/pip (python 3.12)
```

- use pip3 as default pip

```
tasks:
- name: Set pip3 as default pip
  alternatives:
    name: pip
    path: /usr/lib/python3/dist-packages/pip
```

```
TASK [Set pip3 as default pip] *****
fatal: [server2]: FAILED! => {"changed": false, "msg": "Needed to install the alternative, but unable to do so as we are missing the link"}
fatal: [server1]: FAILED! => {"changed": false, "msg": "Needed to install the alternative, but unable to do so as we are missing the link"}
```

This code shows how to set pip3 as default, however, it was not successful and ran into multiple errors even after troubleshooting.

- use python3 as default python

```
- name: Set python3 as default python
  alternatives:
    name: python
    path: /usr/bin/python3
```

This code shows how to set python3 as default, however, it was not successful and ran into multiple errors even after troubleshooting.

- Install Java open-jdk

```
- name: Install OpenJDK
  package:
    name: openjdk-11-jdk
    state: present
```

```
TASK [Install OpenJDK] *****
changed: [server1]
changed: [server2]
```

This code shows how to install Java open-jdk, as shown on the output, it was successful on both manage nodes.

```
jalene@server1:~$ java --version
openjdk 11.0.29 2025-10-21
OpenJDK Runtime Environment (build 11.0.29+7-post-Ubuntu-1ubuntu124.04)
OpenJDK 64-Bit Server VM (build 11.0.29+7-post-Ubuntu-1ubuntu124.04, mixed mode, sharing)
```

```
jalene@server2:~$ java --version
openjdk 11.0.29 2025-10-21
OpenJDK Runtime Environment (build 11.0.29+7-post-Ubuntu-1ubuntu124.04)
OpenJDK 64-Bit Server VM (build 11.0.29+7-post-Ubuntu-1ubuntu124.04, mixed mode, sharing)
```

- Install MariaDB as well as starting the server, create a database and a table using mariaDB and input one record into a table USING ANSIBLE ONLY

```
- name: Install MariaDB
  package:
    name: mariadb-server
    state: present
```

```
TASK [Install MariaDB] *****
ok: [server2]
ok: [server1]
```

```
- name: Start MariaDB Service
  service:
    name: mariadb
    state: started
    enabled: true
```

```
TASK [Start MariaDB Service] *****
ok: [server2]
ok: [server1]
```

```
jalene@server1:~$ sudo systemctl status mariadb
[sudo] password for jalene:
● mariadb.service - MariaDB 10.11.13 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-01-29 08:39:48 UTC; 1h 33min ago
```

```
jalene@server2:~$ sudo systemctl status mariadb
[sudo] password for jalene:
● mariadb.service - MariaDB 10.11.13 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-01-29 08:39:50 UTC; 1h 34min ago
```

The codes shown shows how to install and set the MariaDB server active. It was successful.

```
- name: Create Database
  mysql_db:
    name: "{{ db_name }}"
    state: present
```

```
- name: Create Table
  mysql_query:
    db: "{{ db_name }}"
    query: >
      CREATE TABLE IF NOT EXISTS {{ db_table }}
      (id INT PRIMARY KEY, name VARCHAR(50));
```

```
- name: Insert Record
  mysql_query:
    db: "{{ db_name }}"
    query: >
      INSERT INTO {{ db_table }}
      VALUES ({{ db_record.id }}, '{{ db_record.name }}');
```

```
TASK [Create Database] *****
fatal: [server2]: FAILED! => {"changed": false, "msg": "A MySQL module is required: for Python 2.7 either PyMySQL,
or MySQL-python, or for Python 3.X mysqlclient or PyMySQL. Consider setting ansible_python_interpreter to use the
intended Python version."}
...ignoring
fatal: [server1]: FAILED! => {"changed": false, "msg": "A MySQL module is required: for Python 2.7 either PyMySQL,
or MySQL-python, or for Python 3.X mysqlclient or PyMySQL. Consider setting ansible_python_interpreter to use the
intended Python version."}
...ignoring

TASK [Create Table] *****
fatal: [server1]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'db_n
ame' is undefined. 'db_name' is undefined\n\nThe error appears to be in '/home/jalene/Santiago PrelimExam/playbook
.yml': line 36, column 5, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offendin
g line appears to be:\n\n\n  - name: Create Table\n    ^ here\n"}
fatal: [server2]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'db_n
ame' is undefined. 'db_name' is undefined\n\nThe error appears to be in '/home/jalene/Santiago PrelimExam/playbook
.yml': line 36, column 5, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offendin
g line appears to be:\n\n\n  - name: Create Table\n    ^ here\n"}

PLAY RECAP *****
server1      : ok=6    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=1
server2      : ok=6    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=1
```

For this task, I ran into multiple errors like interpreter issues. I wasn't able to do it correctly.

- Create Motd containing the text defined by a variable defined in config.yaml file and if there is no variable input the default motd is "Ansible Managed node by (your user name)"

```
- name: Set MOTD
  copy:
    content: "{{ motd_message | default('Ansible Managed node by Jalene Santiago' ) }}"
    dest: /etc/motd
```

```
TASK [Set MOTD] *****
changed: [server2]
changed: [server1]
```

This code shows how to create an MOTD, based on the output, it was successfully created.

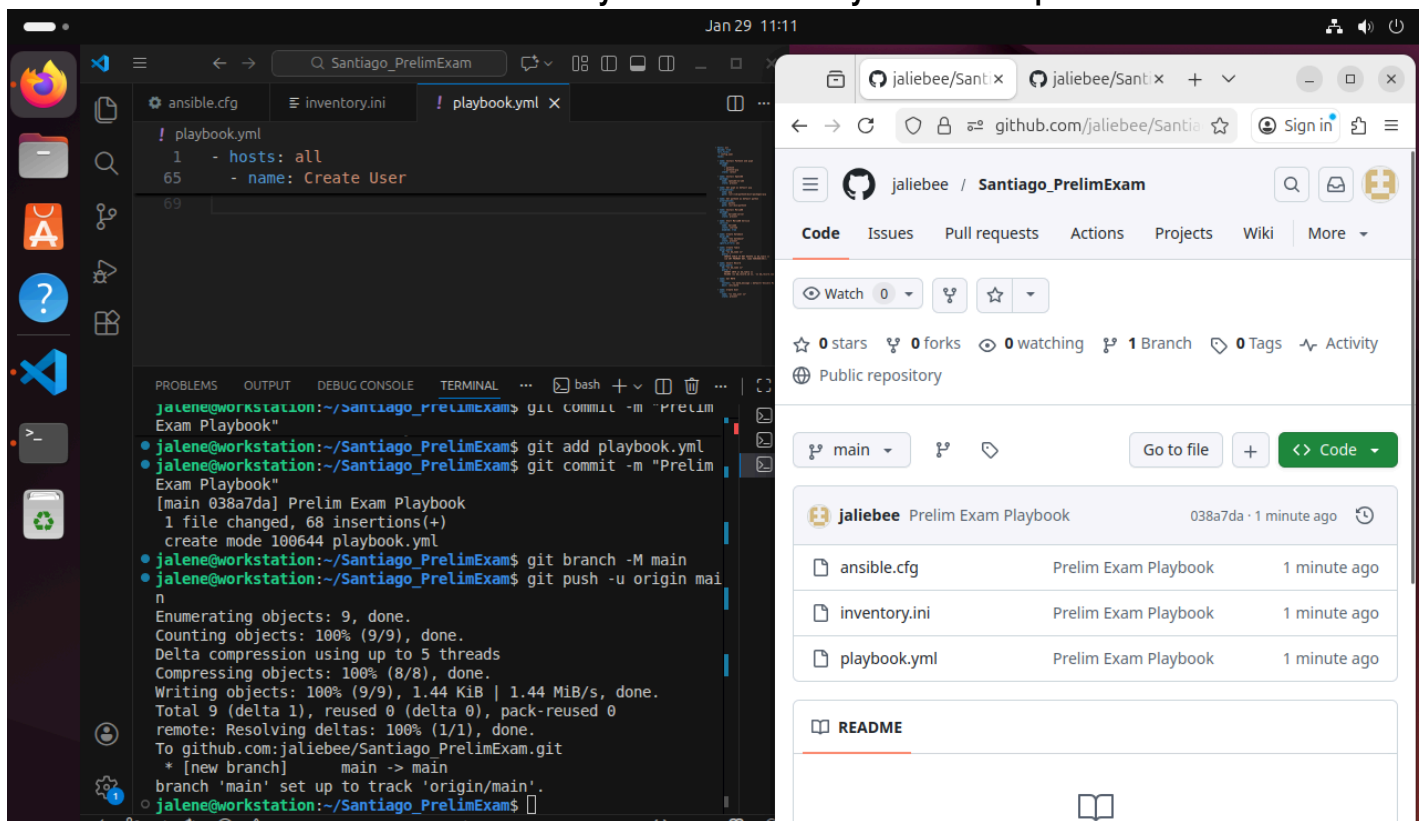
- Create a user with a variable defined in config.yaml

```
- name: Create User
  user:
    name: "{{ new_user }}"
    state: present
```

```
TASK [Create User] *****
fatal: [server1]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'new_user' is undefined. 'new user' is undefined\n\nThe error appears to be in '/home/jalene/Santiago_PrelimExam/playbook.yml': line 7, column 5, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n\n  - name: Create User\n    ^ here\n"}
fatal: [server2]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'new_user' is undefined. 'new user' is undefined\n\nThe error appears to be in '/home/jalene/Santiago_PrelimExam/playbook.yml': line 7, column 5, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n\n  - name: Create User\n    ^ here\n"}
```

This code shows how to create user with a variable defined in config.yaml. However, I couldn't still do it successfully.

PUSH and COMMIT your PrelimExam in your GitHub repo



The screenshot displays a terminal window on the left and a web browser showing a GitHub repository on the right.

Terminal Output:

```
jalene@workstation:~/Santiago_PrelimExam$ git add playbook.yml
jalene@workstation:~/Santiago_PrelimExam$ git commit -m "Prelim Exam Playbook"
[main 038a7da] Prelim Exam Playbook
1 file changed, 68 insertions(+)
create mode 100644 playbook.yml
jalene@workstation:~/Santiago_PrelimExam$ git branch -M main
jalene@workstation:~/Santiago_PrelimExam$ git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 5 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 1.44 KiB | 1.44 MiB/s, done.
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
To github.com:jaliebee/Santiago_PrelimExam.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
jalene@workstation:~/Santiago_PrelimExam$
```

GitHub Repository Page:

The page shows the repository **jaliebee / Santiago_PrelimExam**. It includes a table of files:

File Name	Commit Hash	Time
ansible.cfg	038a7da	1 minute ago
inventory.ini	038a7da	1 minute ago
playbook.yml	038a7da	1 minute ago

GitHub Link:

github.com/jaliebee/Santiago_PrelimExam